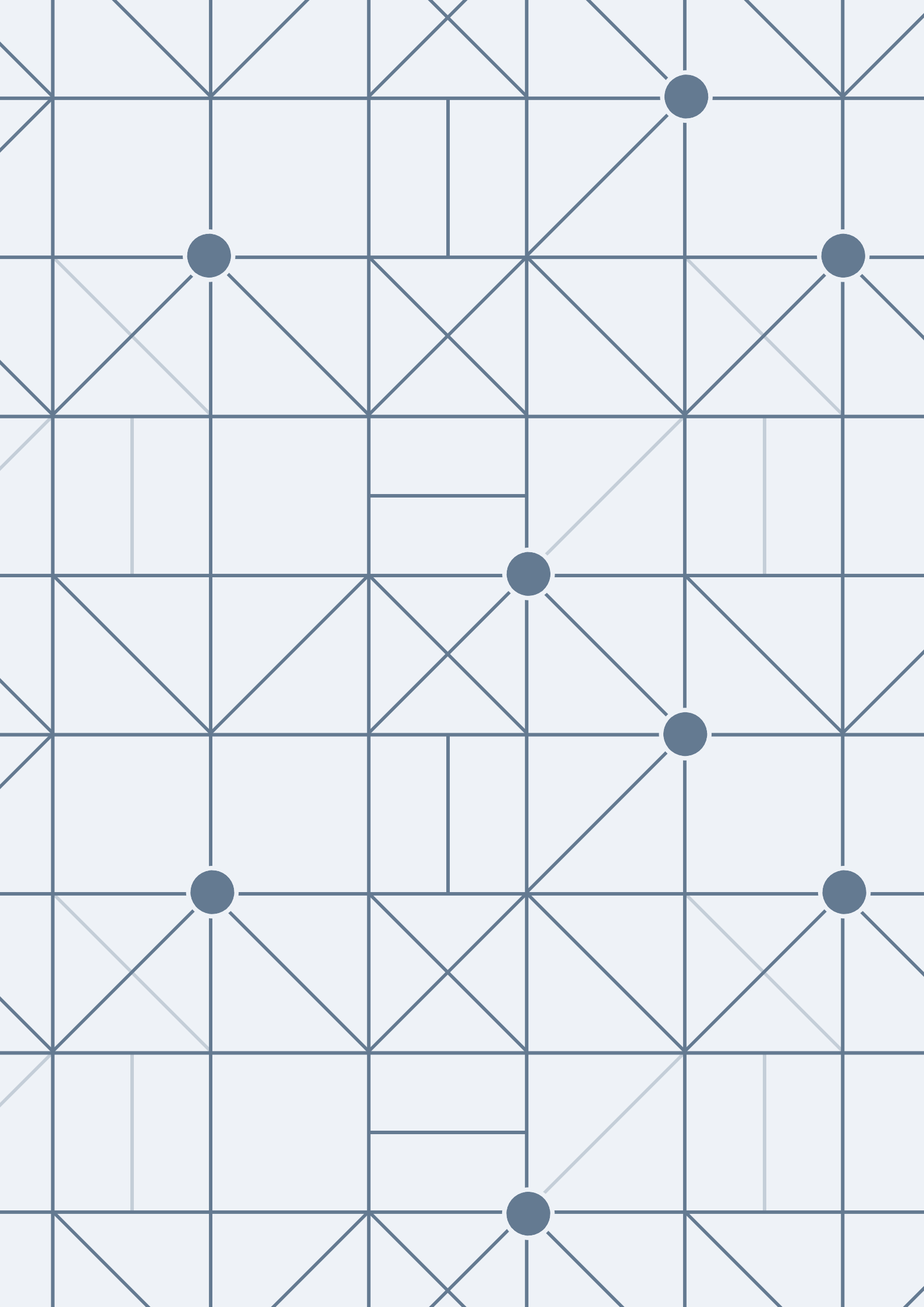




Temporal Evolution of a Microstate

Classical mechanics

$H(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + V(\mathbf{q})$

Hamiltonian equations of motion

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}$$

Initial state: $\mathbf{q}(0), \mathbf{p}(0)$

Final state: $\mathbf{q}(t), \mathbf{p}(t)$

$\forall i = 1, \dots, s$

- Trajectories cannot leave the accessible phase space. Two or more trajectories **never intersect each other** (the laws of **Mechanics are deterministic**)
- If two systems are characterized by the same Hamiltonian H and are in the same initial microstate (q_0, p_0) , after a certain time t , both will be in the **same final microstate**:
 $(q_0, p_0) = (q'_0, p'_0) \Rightarrow (q_t, p_t) = (q'_t, p'_t)$
- If at $t = 0$, one system is in a microstate (q_0, p_0) , and another system with the same Hamiltonian is in a different one (q'_0, p'_0) , then after a time t , the two will be in **different microstates**:
 $(q_0, p_0) \neq (q'_0, p'_0) \Rightarrow (q_t, p_t) \neq (q'_t, p'_t)$
- Classic: two systems with the same Hamiltonian prepared in two very similar microstates evolve microscopically in very different ways, **high sensitivity to initial conditions**.

$\mathcal{H}(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + V(\mathbf{q})$
 $\vec{F} = \langle p_1, p_2, \dots, p_s \rangle$
 $\vec{q} = \langle q_1, q_2, \dots, q_s \rangle$

Let's consider a **dynamic function** $a(\mathbf{q}, \mathbf{p})$ that doesn't explicitly depend on time. Despite its lack of explicit time dependence, this function can change over time as a result of the system's motion, given that $q_i = q_i(t)$ and $p_i = p_i(t)$

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial a}{\partial p_i} \frac{dp_i}{dt} \right) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \dot{q}_i + \frac{\partial a}{\partial p_i} \dot{p}_i \right)$$

and if we use the Hamiltonian equations:

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial a}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \equiv \{a, H\}$$

Poisson brackets

If the dynamic function depends explicitly on time $\rightarrow \frac{d}{dt}a(\mathbf{q}, \mathbf{p}, t) = \{a, H\} + \frac{\partial a}{\partial t}$

Per my dynamic function $a = a(\mathbf{q}, \mathbf{p})$

$\dot{a} = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \dot{q}_i + \frac{\partial a}{\partial p_i} \dot{p}_i \right) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{\partial H}{\partial p_i} + \frac{\partial a}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = \{a, H\}$

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \frac{da}{dt} = \{a, H\}$$

This equation is the most important equation in Classical Mechanics and includes the canonical equations of Hamilton as a particular case

$$\dot{q}_i = \{q_i, H\} = \sum_{j=1}^s \left(\frac{\partial q_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial q_i}{\partial p_j} \frac{\partial H}{\partial q_j} \right) = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = \{p_i, H\} = \sum_{j=1}^s \left(\frac{\partial p_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial p_i}{\partial p_j} \frac{\partial H}{\partial q_j} \right) = -\frac{\partial H}{\partial q_i}$$

$$\dot{H} = \{H, H\} = 0$$

H is basically a constant of motion. In general, if a certain dynamic function a satisfies the condition $\{a, H\} = 0$, then $a(\mathbf{q}, \mathbf{p})$ is a **constant of motion** of the system.

QUANTUM DESCRIPTION

Uncertainty Principle

An important difference between Classical Mechanics and Quantum Mechanics is established by the **Uncertainty Principle**.

"The position and momentum of a particle cannot be simultaneously determined in a measurement with arbitrary precision."

$$\Delta q_i \cdot \Delta p_i \geq \frac{\hbar}{2} \quad \text{for all } i = 1, \dots, s$$

Planck constant

$\hbar = \frac{h}{2\pi} \approx 1.0546 \times 10^{-34} \text{ Js}$

$\hbar = 6.62 \times 10^{-34} \text{ Js}$

CONSEQUENCE

A quantum microstate **cannot** be projected as a **point in phase space**. we must represent it as cells with dimensions $\Delta q_i \Delta p_i$ that satisfy the Heisenberg Principle. All points within a cell are quantum mechanically **indistinguishable** and therefore **correspond to the same quantum state**.

Note: The uncertainty Principle is **not** a limitation on our ability to measure accurately but reflects a **fundamental property** of quantum systems

$$\Delta q_i \cdot \Delta p_i \geq \frac{\hbar}{2} \quad \text{for all } i = 1, \dots, s$$

The cells, which are regions in phase space rather than single points, provide a **probabilistic description** of a quantum microstate.

This quantum description leads to a more probabilistic and statistical understanding of physical systems, fundamentally different from classical mechanics.

When should a problem be considered quantum or classical?

Hypervolume of uncertainty

$\hbar_s/2$

A problem must be addressed by **Quantum Mechanics** when the **action of the system is on the order of the Planck constant**

$$\mathcal{S} = \int_{q_1}^{q_2} \mathcal{L} dt \approx \hbar$$

Classical treatment will only be correct in the limit where the **action is much greater than the Planck constant**, allowing it to be considered that $\hbar$ is negligible

$$\mathcal{S} = \int_{q_1}^{q_2} \mathcal{L} dt \gg \hbar$$

$\mathcal{L} = T - V$
Lagrangian

The smallest possible action is $\hbar/2$; larger action values are integer multiples of this quantum

Quantum or Classical Mechanics?

Ehrenfest's Theorem

Ehrenfest's Theorem states that a wave packet can be treated using Hamilton's equations (i.e., as a classical particle) if:

- $\Delta x \ll d$, where d is the typical distance traveled;
- $\Delta p_x \ll \bar{p}$, average momentum of the particle (wave packet);
- $\hbar/2 \leq \Delta x \Delta p_x \ll d \cdot \bar{p} \Rightarrow d \gg \hbar/2p$

At **high temperatures**, thermal agitation causes the particles of a system (molecules, atoms, etc.) to move over **long distances** ($d \gg 0$) and possess **very large linear momenta** ($p \gg 0$). In other words, $d \gg \hbar/2p$, and we can apply the classical treatment.

At **low temperatures**, just the opposite happens: $d < \hbar/2p$, and it's necessary to apply QM.

^a In the formulation of QM, the state of a system is represented by the **state vector** $|\Psi\rangle$, which encodes the **complete quantum state** of a physical system, including details about the positions, momenta, and other properties of its particles or components.

^b $|\Psi\rangle$ serves as a mathematical representation that **encodes predictions about the probabilities** of various outcomes when measurements are performed on the system.

^c At a particular instant of time, the components of this vector represent the values of the corresponding **wave function**, which is a mathematical representation of the state vector (and hence of the quantum state of the system) in a particular basis, typically the position or momentum basis:

$$\Psi = \Psi(q_1, \dots, q_n, t)$$

$$|\Psi(t)\rangle = \int \Psi(x, t) |x\rangle dx$$

Position state vector = wave function $\Psi(x, t)$
 (and not Dirac's notation)

Source: <https://www.youtube.com/watch?v=UW3333333333>

state vector $|\Psi\rangle$ encodes the complete state
 but we do not need all the information, if we are interested in the position
 wave function $\Psi(x, t)$ (position space)